IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

**SMART GRID AND DATA SCIENCE**

Friday, 9 May 2025, 10:00 AM

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions.**

*All questions carry equal marks.*

**Any special instructions for invigilators and information for candidates are on page 1.**

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Second Marker(s) : F. Teng, G. Strbac, G. Strbac

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## The Questions

**Question 1 (20 marks)**

a. Describe the pros and cons of preventive and corrective modes that improve system security.

[3]

b. Wakanda's power system, presented in Figure 1.1, includes a 1.5 GW gas-turbine generator in the South with a marginal cost of £60/MWh and a 300 MW peaking generator in the North-West with a marginal cost of £120/MWh. The loads are 200MW in the South, 750MW in the North-West, and 500MW in the North-East.

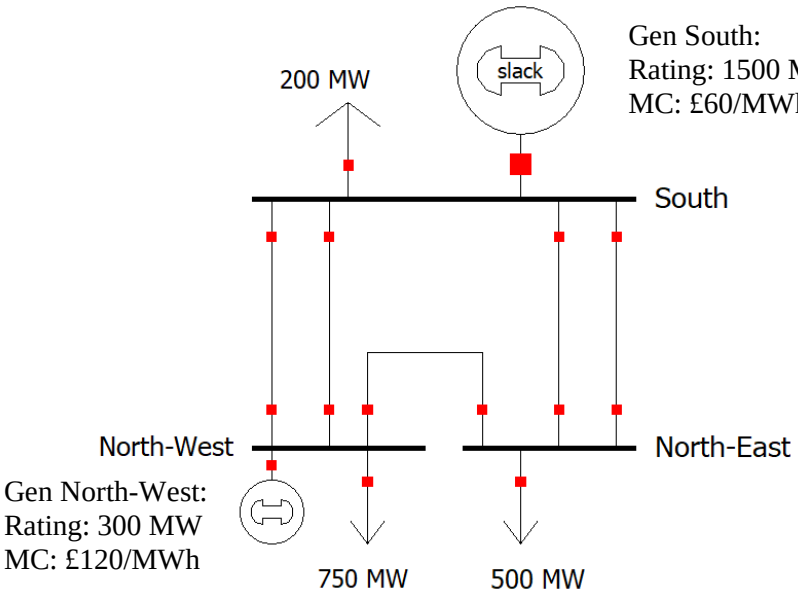



**Figure 1.1: Wakanda's power system**

Table 1.1 below shows the parameters of the transmission corridors connecting the North and South regions and between the North-East and the North-West regions.

**Table 1.1: Parameters of transmission corridors**

| Transmission corridor | Number of circuits | X (p.u.) | Capacity per circuit (MW) |
|---|---|---|---|
| North-East – South | 2 | 0.50 | 500 |
| North-West – South | 2 | 1.00 | 375 |
| North-East – North-West | 1 | 0.25 | 500 |

Answer the following questions:

i) Determine the Economic Dispatch and compute the corresponding power flows in each of the circuits

[2]

ii) Determine the power flows following the outage of an individual single circuit and identify conditions in which the power flows during the contingency exceed the circuit capacity limit.

[3]

iii) If Demand Side Response is available at North-East to provide corrective control, please determine the minimum amount of Demand Side Response needed to manage the power flows within limits.

[6]

iv) If the system operator needs to secure the system in a preventive mode against a line outage using the North-West generator, determine the secured dispatch and the corresponding cost. What is the value of Demand Side Response at North-East per MW made available if the probability for that line outage is 1%?

[6]

**Question 2 (20 marks)**

a) A grid-connected Battery Energy Storage System (BESS) has the following specifications: Capacity: 100 MWh; Maximum charge/discharge rate: 20 MW. Assume the battery is fully charged at the start of the peak period.

i) How long can the BESS supply power at its maximum discharge rate before it needs to be recharged?

[2]

ii) During this discharge period, how much energy (in MWh) is delivered to the grid, if accounting for the storage discharging efficiency at 90%?

[2]

iii) If the BESS is recharged for 5 hours with a charging power of 15 MW accounting for the storage charging efficiency at 90%, what percentage of the battery's capacity will be recharged?

[2]

b) A BESS has a capacity of 200 MWh and can charge and discharge at a maximum rate of 100 MW. Given the electricity price as shown in Figure 2.1 and the initial state-of-the charging (SoC) at 50%, calculate and draw the battery's daily charging and discharging power, with the objective of maximising the energy arbitrary (revenue). Calculate the daily revenue of the BESS.

[4]

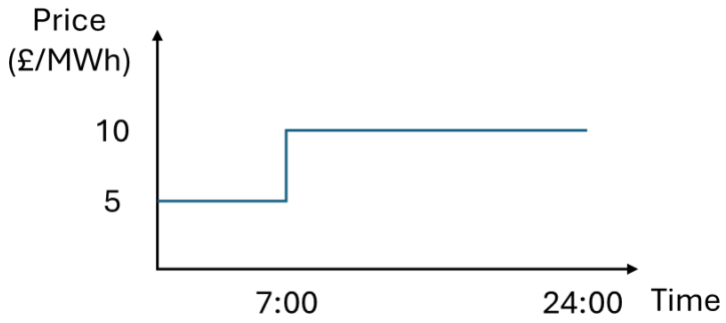



**Figure 2.1. Daily electricity price.**

c) Following the setup in b), calculate and draw the battery's daily charging and discharging power when considering the charging and discharging efficiencies at 90%. Calculate the daily revenue of the BESS.

[8]

d) Following the setup in c), assume that the daily eletricity price profile is consistent across the year and the annualised cost of BESS system is 450k, is it worth to invest?

[2]

**Question 3 (20 marks)**

a) Monte Carlo simulation is applied to estimate the expected number of interruptions.

i) The application of Monte Carlo simulations relies on three fundamental theories. Please state these three theories and explain how they support the application.

[3]

ii) After N=10,000 samples the best estimate of the interruption is 110,000 MW, with a standard error of 2093 MW. How many samples will be required (approximately) to reduce the coefficient of variation to 0.52%?

[3]

b) Give a brief definition for big data and mention the primary sources of big data in power systems.

[3]

c) Describe how dynamic security assessment has been carried out traditionally and how it can be improved by using Phase Measurement Unit (PMU) data and machine learning methods

[4]

d) For the distribution network with high penetration of smart meter at the household level,

i) State examples of key information can be identified through smart meter data.

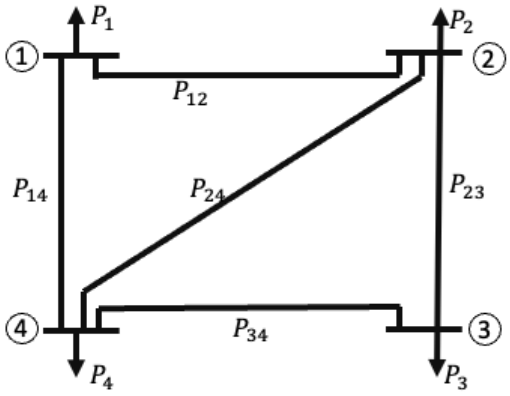
[3]

ii) List at least three Privacy Enhancing Technologies (PETs) and discuss their suitability for smart metering system.

[4]

**Question 4 (20 marks)**

Consider the following power system with 4 buses and 5 branches. The branch reactance and power flow measurements are given by the tables below. The per unit base is given as 100MVA. Bus 1 is set as the slack bus. *Note: By convention, $\boldsymbol{P_{12}}$ represents the power flow from bus 1 to bus 2.*




**Line Reactance (p.u.)**

| | |
|---|---|
| $x_{12}$ | 0.4 |
| $x_{14}$ | 0.1 |
| $x_{23}$ | 0.2 |
| $x_{24}$ | 0.2 |
| $x_{34}$ | 0.5 |

**Measurement (MW)**

| | |
|---|---|
| $P_{12}$ | 50 |
| $P_{13}$ | 500 |
| $P_{14}$ | -50 |
| $P_{24}$ | 150 |
| $P_{34}$ | 80 |

**Figure 4.1 4-bus system with branch reactance and measurements**

a) The incidence matrix of above power system can be written as follows. According to the incidence matrix, construct the measurement matrix **H** for this system under the DC assumptions. What is the rank of **H** and is the system observable and why?

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

[5]

b) Find the power injection on each bus.

[2]

c) Given the power flow measurements and the measurement matrix **H** in a), find the current state of the system and the residual of the measurement.

[3]

d) Given that the load flow limit on line 1-3 is 1000MW, an attacker would like to convince the system operator that line 1-3 is overloaded, e.g. to modify the measurement $\mathbf{P_{13}}$ from 500MW to 1000MW. Is it possible to only change $\mathbf{P_{13}}$ without being detected by the bad data detection (BDD) and why? Assume the attacker only changes the voltage angle at bus 4, calculate the attack vector that can bypass the BDD under the noiseless condition.

[5]

e) Discuss how measurement noise may affect the detection of False Data Injection Attacks (FDIA)

[2]

f) Briefly describe the concept of Moving Target Defence (MTD) against False Data Injection Attacks (FDIA). Based on the attack formulated in iv), determine which transmission line reactance should be modified if only one can be changed.

[3]